

SMART DISPLAY MODULE SPECIFICATION

1.5 Inch Smart Knob Display with Touch	
Model:	UEDX46460015-MD50ET
Version:	V2.0
Date:	2026-04-01

Customer Confirmation

Approved by	Notes

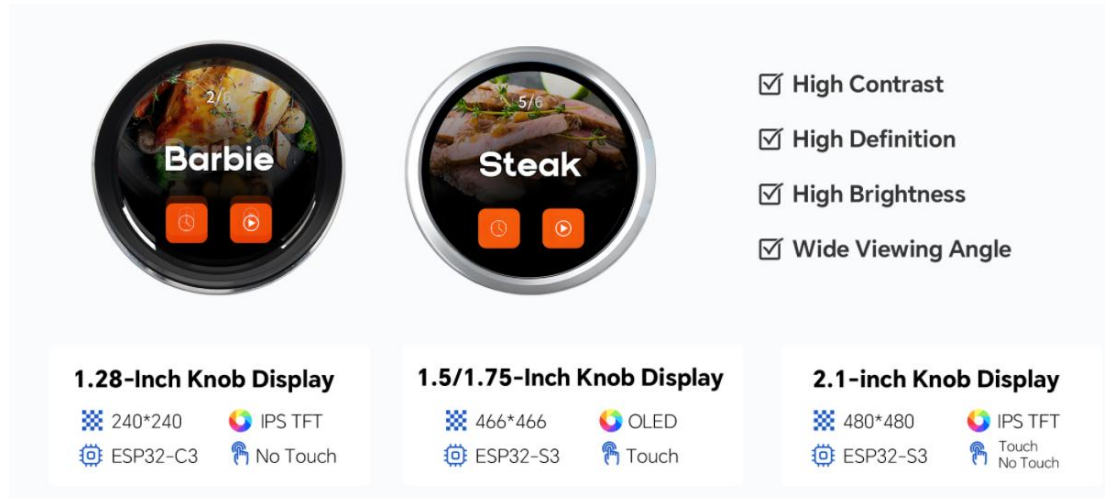
REVISION HISTORY

Revision	Date	Contents of Revision Change	Remark
V1.0	20241010	Preliminary release	
V1.1	20241016	Add GitHub project links, and environment config links	
V1.2	20250710	Increase the correspondence between adapter pins and ESP32 pins	
V2.0	20260401	Update mechanical drawings, promotional images and materials	

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1. Introduction



1.1 Features

Brief Info:

- 1) Outline Dimension: ϕ 51.6 Round
- 2) Interaction Method: Rotate and Press
- 3) Shell Color: Black/Silver/Customized
- 4) Power: DC 5V, 1A

System

- 1) OS: RTOS
- 2) CPU: ESP32-S3 240Mhz
- 3) RAM: 8MB
- 4) Flash: 16MB
- 5) Interface: UART/USB
- 6) Support 2.4GHz Wi-Fi, BLE 5, BLE Mesh

Note: The Bluetooth and WIFI functions are still under test, we will complete them as soon as possible, please forgive the inconvenience caused to you

Display

- 1) Size: 1.5 Inch
- 2) Resolution: 466*466
- 3) Mode: AMOLED
- 4) Driver IC: CO5300AF-42
- 5) Interface type: QSPI
- 6) Brightness: 1000 cd/m²
- 7) Touch IC: CST820

Other

- 1) Operation Temperature: -20~60°C
- 2) Storage Temperature: -30~70°C

2. Product information

2.1 Connector Interface Description

Pin NO.	Symbol	Description	Voltage Range	Remarks
1	VCC	Power 5V	5V	
2	GPIO38	-	-	
3	GND	Grounds	0V	
4	GPIO40	GPIO40	-	
5	GPIO39	GPIO39	-	
6	RX	UART Receive	0-3.3V	
7	TX	UART Transmit	0-3.3V	
8	CHIP-EN	CHIP-EN	0-3.3V	
9	D+	USB D+	3.3V	
10	D-	USB D-	3.3V	

The connector specifications is 10PIN 0.5mm pitch

10pin FPC:

FPC number	adapter pin	ESP32S3 Pin
1	5V	5V
2	PB7	-
3	GND	GND
4	RX2	GPIO40
5	TX2	GPIO39
6	RX1	RXD/GPIO44
7	TX1	TXD/GPIO43
8	-	CHIP-EN
9	SK & D+	USB D+/GPIO20
10	SD & D-	USB D-/GPIO19

2.2 Display interface description

Pin No.	Symbol	I/O	Description
1	VCI_EN	P	Power supply
2	NC	-	-
3	NC	-	-
4	NC	-	-
5	QSPI_SI03	I	QSPI Data3 input pin
6	QSPI_SI02	I	QSPI Data2 input pin
6	QSPI_SI01	I	QSPI Data1 input pin
7	QSPI_SI00	I/O	QSPI Data0 input pin
9	NC	-	-
10	SCL	I	SPI pclk input signal
11	NC	-	-
12	CS	I	Chip selection pin. Low-active
13	GND	P	Power Ground
14	TE	O	Tearing effect signal is used to synchronize MCU to frame memory
15	IOVCC	P	Power supply to the internal logic power regulator
16	MTP(NC)	-	-
17	VBAT	P	Power supply
18	RST	I	The signal will reset the LCM, Signal is active low.
19	TP_SCL	I	I2C clock signals for CTP
20	TP_SDA	I/O	I2C data signal for CTP
21	TP_RST	I	The signal will reset the CTP, Signal is active low
22	TP_INT	I	Interrupt signals for CTP

23	TP_VCC	P	Power supply for TP
24	GND	P	Power Ground

2.3 Display Information

Item	Specification	Unit
Pixel Driving element	AMOLED	-
Screen Size	1.51	Inch
Resolution	466*466	Dots
Module Power Consumption	-	Watt
Active Area	38.3052*38.3052	mm
Pixel pitch (W*H)	82.2	mm
Module Size (W*H*D)	44.3(W)*44.3(H)*2.405(D)	mm
Driver IC	CO5300AF-42	-
Interface Mode	QSPI	-
Display mode	Normal Black	-
Luminance	450	cd/m ²
Viewing Direction	ALL	O'clock
Display Color	16.7M	Colors
LED life time	-	-

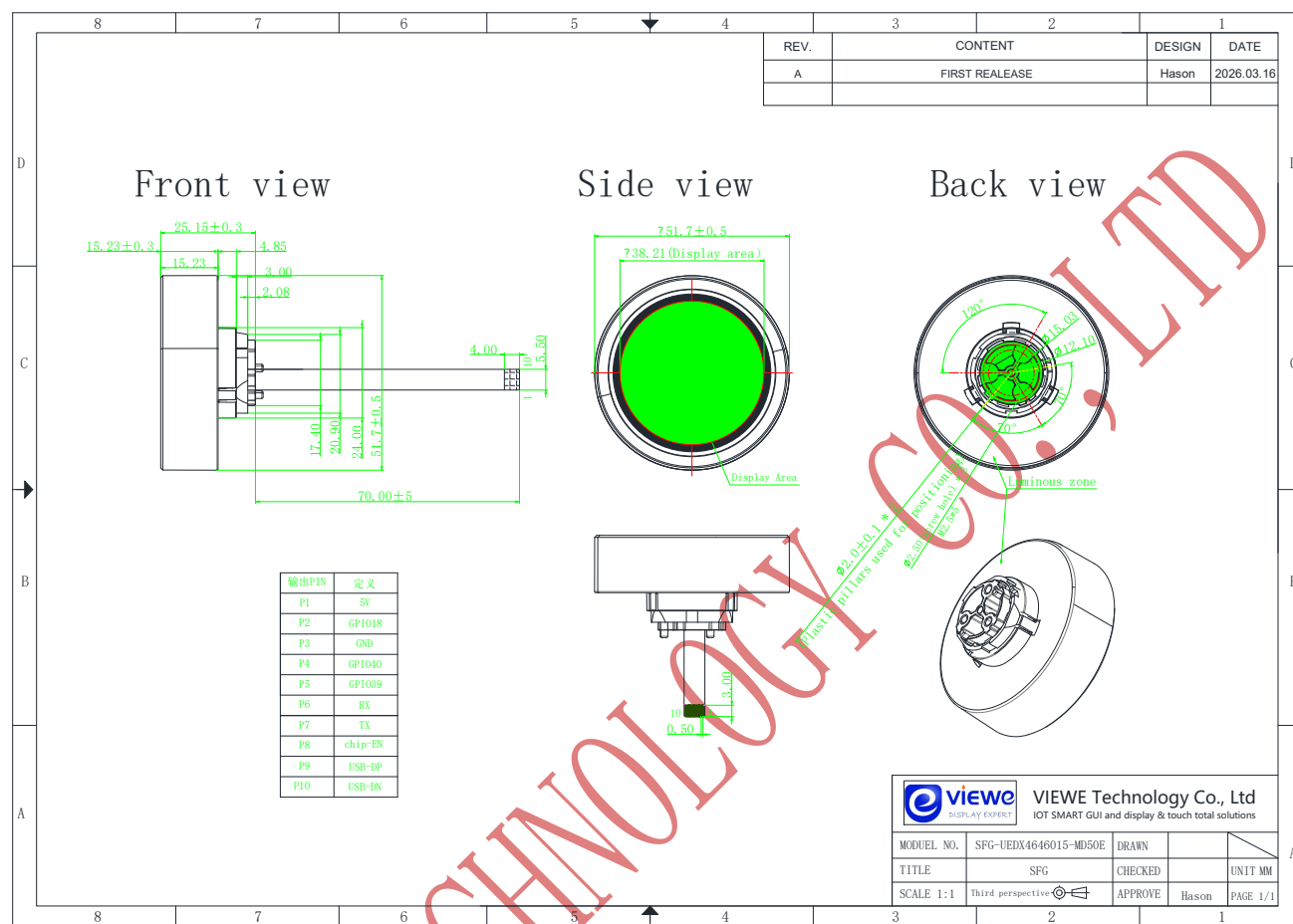
2.4 Voltage & Current

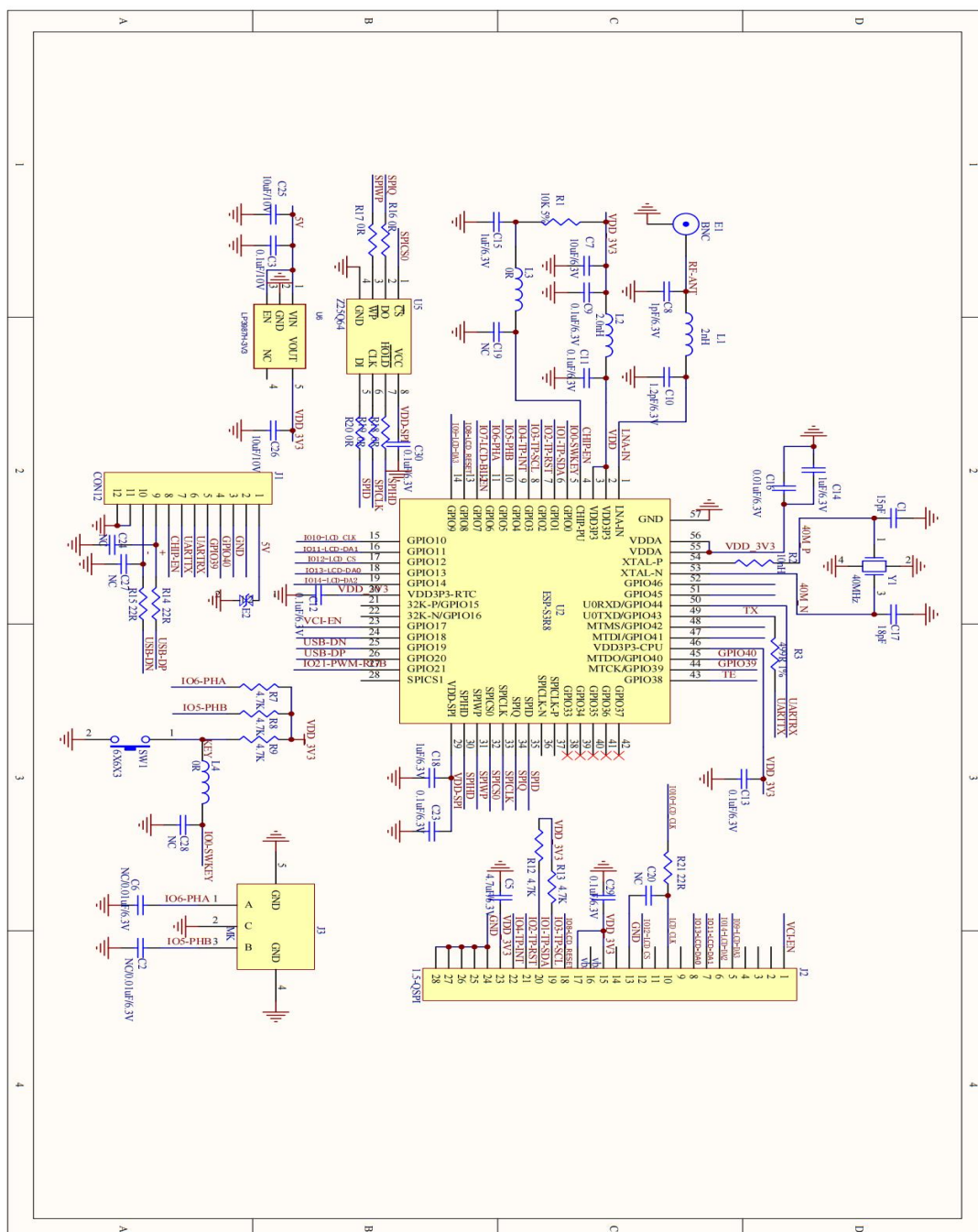
Item	Conditions	Min	Typ	Max	Unit
Power Voltage	DC	4.0	5.0	5.5	V
Operation Current	VCC= +5V, Maximum backlight current	50	100	150	mA
	VCC= +5V, backlight off	-	50	-	mA
Recommended power supply: 5V 1A DC					

2.5 Reliability Test

Item	Conditions	Min	Typ	Max	Unit
Working Temperature	60%RH at 5V voltage	-20	25	60	C
Storage Temperature	---	-30	25	70	C
Working Humidity	25°C	10%	60%	90%	RH
ESD	---	Contact: ±4KV Air: ±8KV			KV

3. MECHANICAL DRAWING





5. Related downloads

5.1 Arduino and IDF relevant information

<https://github.com/VIEWESMART/UEDX46460015-MD50ESP32-1.5inch-Touch-Knob-Display>

5.2 Libraries required for Arduino

<https://github.com/VIEWESMART/UEDX46460015-MD50ESP32-1.5inch-Touch-Knob-Display/tree/main/Libraries>

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